

Smart City Energy Optimization & Prediction Platform

Presenter: Team 4

Nagat Osama | Merna Sameh | Ebrahim Ateff | Abdullah Mohamed

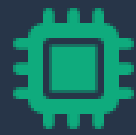
Supervised by: Eng. Ahmed Ali

Project Vision



The Problem

Modern cities face unpredictable energy spikes due to rapid growth that costs cities millions annually. Weather conditions and traffic congestion significantly impact grid instability, leading to inefficiencies.



The Solution

An automated Azure platform integrating Energy, Weather, and Traffic data. Utilizing Medallion Architecture (Raw → Bronze → Silver → Gold) for robust processing.



Unique Value

Proactive Optimization: Not just reporting, but identifying "Risk Zones" and providing actionable load-shifting recommendations before outages occur.

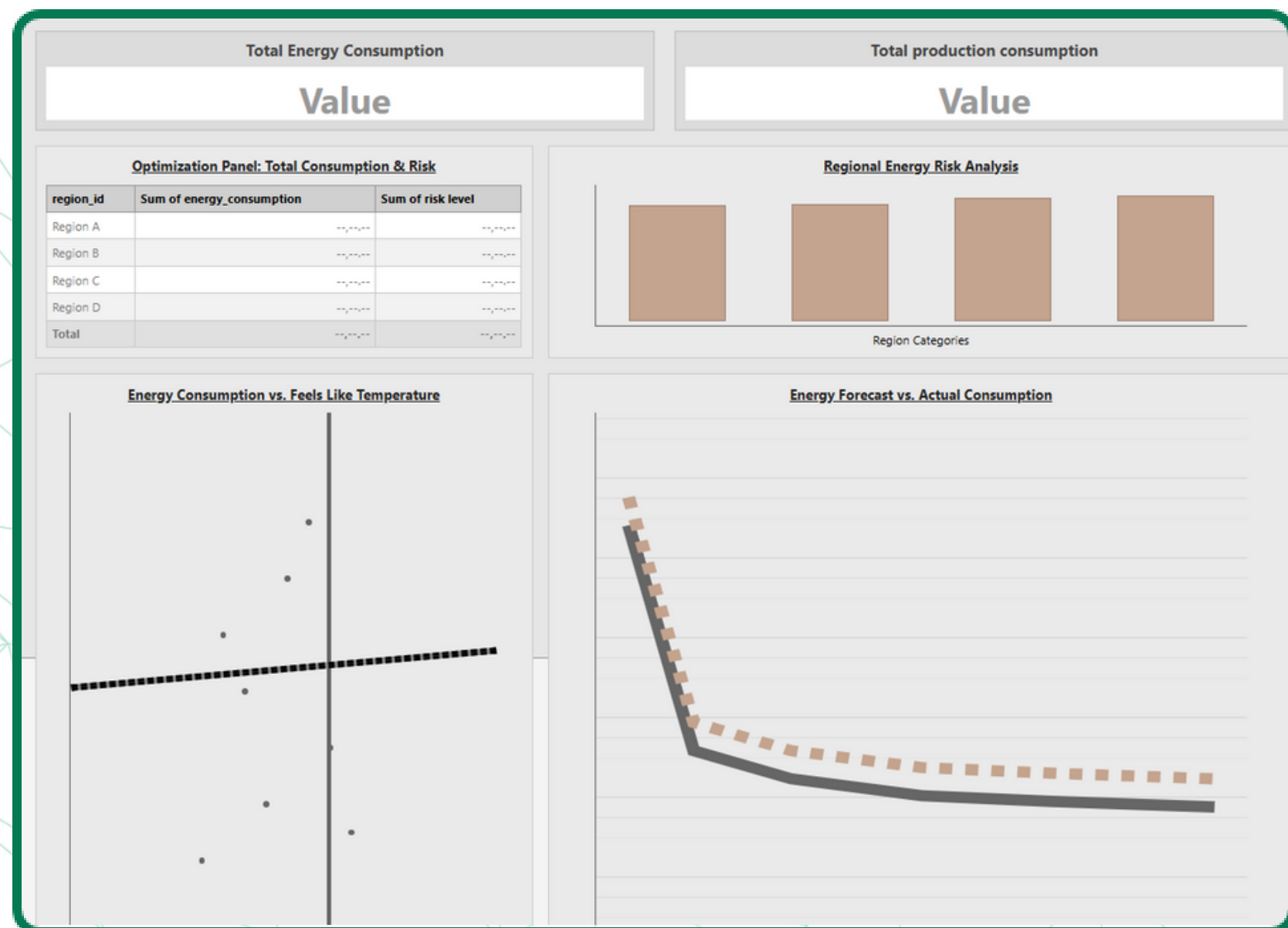
Project Idea



A cloud-native "End-to-End Data Platform" on Azure that integrates batch energy data with real-time weather/traffic APIs to accurately forecast consumption.



Project Wireframe & User Journey



User Journey

- 1 Monitor:** Grid Operator logs in to view real-time grid status and total consumption vs. production.
- 2 Detect:** System flags "High Risk Zones" (Red) where predicted demand > capacity.
- 3 Optimize:** User reviews the "Optimization Panel" for specific actions (e.g., "Shift Load in Region R3").

Users & Core Features

Target Audience

City Planners: Need long-term trends to plan infrastructure expansion and sustainability goals.

Grid Operators: Need granular, hourly forecasts to manage immediate load stability and prevent outages.

Key Features

- ⚡ **Energy Forecasting:** Predicts consumption with high precision using Azure ML (ARIMA/Prophet).
- ⚠ **Risk Identification:** Automatically identifies regions exceeding historical peak capacity.
- ⚖ **Optimization Engine:** Suggests resource balancing strategies (peak shifting) to mitigate risks.

System Architecture



Data Sources: Integration of Batch CSVs and Real-time External APIs (Weather, Traffic).



Ingestion & Processing: Azure Data Factory for orchestration and Databricks for transformation.



AI Modeling: Azure ML Designer for forecasting and risk classification.



Visualization: Synapse SQL Views powering interactive Power BI dashboards.

Medallion Data Architecture

Raw Layer



Stores original data exactly as received (CSVs) for audit and backup.

Bronze Layer



Standardized data with added metadata columns (Load Date, Source System).

Silver Layer



Cleaned, de-duplicated data with feature engineering (Weather Index, Flags).

Gold Layer



Aggregated, business-ready data (Forecasts, Risk Scores) for BI & Analytics.

Data Structure: Medallion Architecture

Key Entities

Energy: Timestamp, Meter_ID,

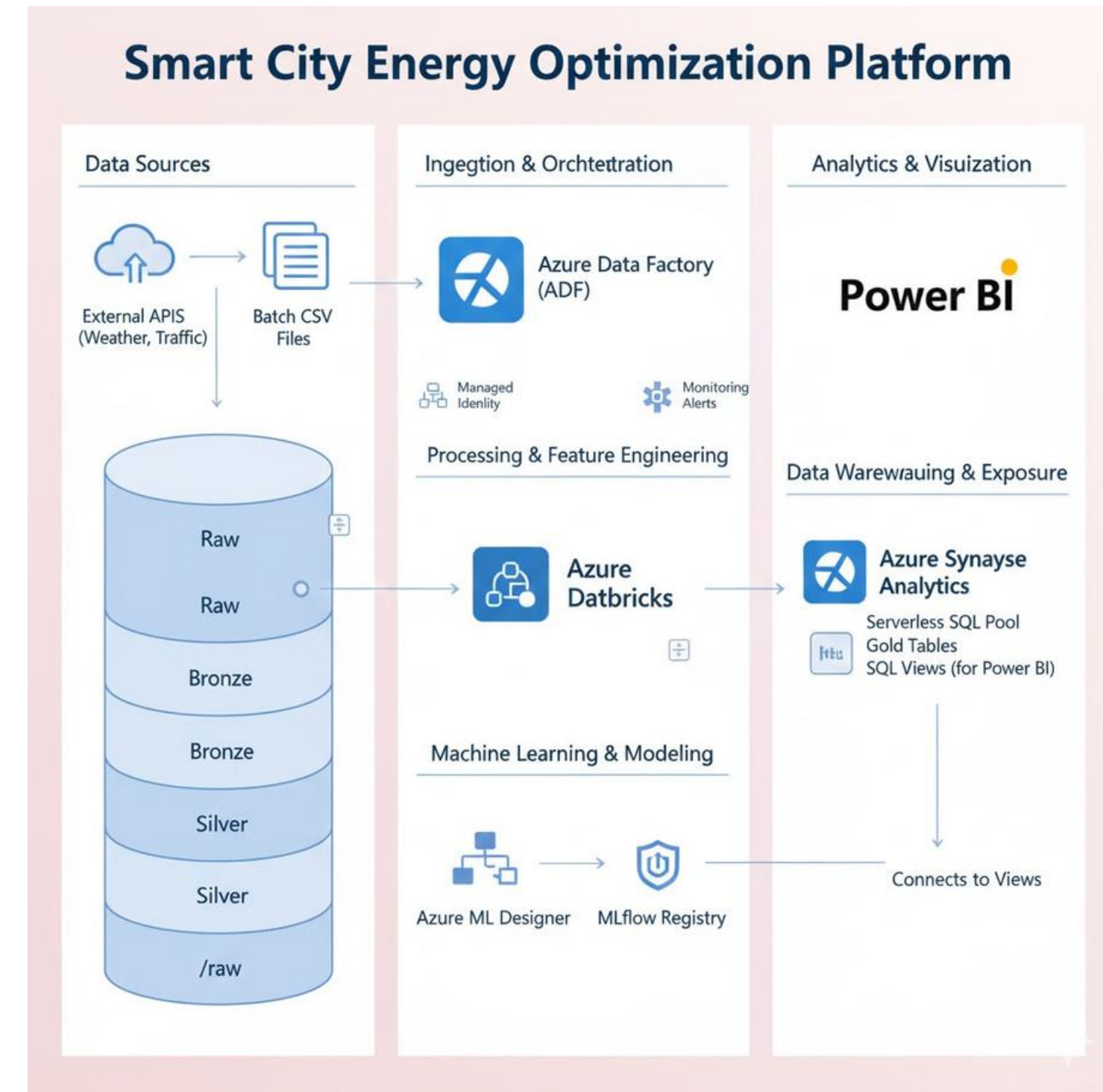
Consumption (kWh).

Weather: Temp, Humidity,

Wind Speed.

Traffic: Congestion Index,

Speed.



Languages & Frameworks



Languages

Python (PySpark): Used in Databricks for complex ETL and Feature Engineering.

SQL: Used in Synapse Serverless Pools for creating reporting views.



Core Frameworks

Azure Data Factory: Orchestration pipelines.

Azure Databricks: Big Data processing.

Delta Lake: ACID transactions.



AI & Analytics

Azure ML Designer: Visual ML pipelines.

MLflow: Model lifecycle management.

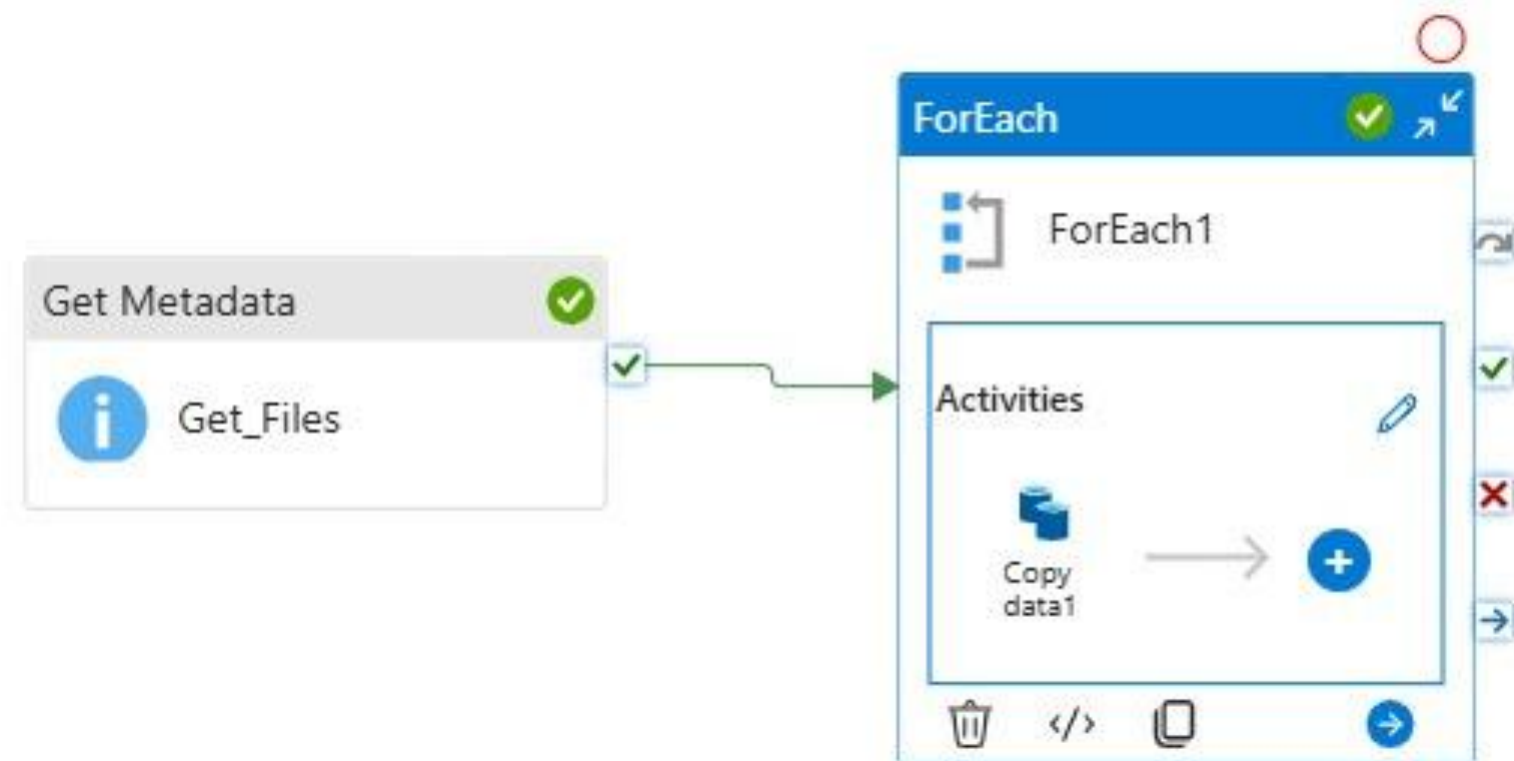
Power BI: Interactive visualization.

Ingestion Pipelines

Presenter: Nagat Osama



Pipeline 1 (Energy): Ingests daily batch CSV files into ADLS Raw and promotes to Bronze.



Ingestion Pipelines

Presenter: Nagat Osama

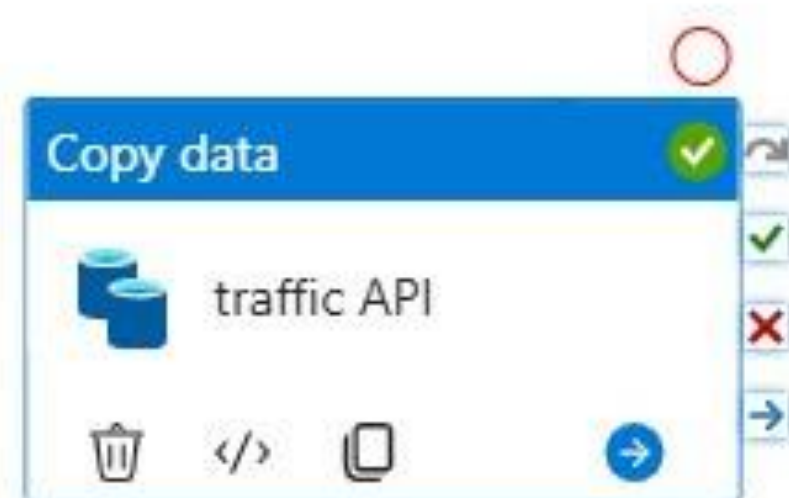
 **Pipeline 2 (Weather):** Fetches real-time REST API data, handling JSON responses.



Ingestion Pipelines

Presenter: Nagat Osama

 **Pipeline 3 (Traffic):** Captures live congestion data via API for correlating load spikes.



Ingestion Pipelines

Presenter: Nagat Osama



hourly
triggers

Add triggers

Choose trigger...

Trigger	Type	Parameters
traffic	Schedule	0

Add triggers

Choose trigger...

Trigger	Type	Parameters
Daily_Weather_Trigg...	Schedule	0

Add triggers

Choose trigger...

Trigger	Type	Parameters
Daily_Files_Trigger	Schedule	0

Ingestion Pipelines

Presenter: Nagat Osama

Would you like to see Data Factory inside of Microsoft Fabric, Microsoft's newest cloud-first data analytics SaaS platform? Click [here](#) to get started with Fabric Data Factory!

Triggered Debug Rerun Cancel options Refresh Edit columns List Gantt

Filter by run ID or name Cairo (UTC+2) : Last 30 days Pipeline name : All Status : All Runs : Latest runs Triggered by : All Add filter Copy filters Export to CSV

Showing 1 - 81 items Last refreshed over 15 minutes ago

☐	Pipeline name ↑↓	Run start ↑↓	Run end ↑↓	Duration	Triggered by	Status ↑↓	Run	Parameters	Annotations	Run ID
☐	pipeline1	11/23/2025, 5:56:06 AM	11/23/2025, 5:56:08 AM	3s	Manual trigger	Failed	Original			7f5144e7-7fed-47ca-9481-d1af5d59f
☐	pipeline1	11/23/2025, 12:47:00 PM	11/23/2025, 12:48:14 PM	1m 14s	Daily_Files_Trigger	Succeeded	Original			5f8f0f66-955d-45a7-8a6e-612a5335c
☐	pipeline1	11/23/2025, 1:47:01 PM	11/23/2025, 1:48:11 PM	1m 10s	Daily_Files_Trigger	Succeeded	Original			38747feb-cc98-4838-bb20-bcd706ae
☐	pipeline1	11/23/2025, 2:47:01 PM	11/23/2025, 2:48:16 PM	1m 16s	Daily_Files_Trigger	Succeeded	Original			91fde051-6be7-4437-b0ed-48665fbb
☐	pipeline1	11/23/2025, 3:47:01 PM	11/23/2025, 3:48:13 PM	1m 13s	Daily_Files_Trigger	Succeeded	Original			1656f469-82da-4b60-9ab5-d95856ff
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☐	pipeline1	11/23/2025, 7:47:00 PM	11/23/2025, 7:48:25 PM	1m 25s	Daily_Files_Trigger	Succeeded	Original			1fa4ea95-c417-438f-b957-fb826cabf
☐	Ingest_Traffic_API	11/23/2025, 8:18:01 PM	11/23/2025, 8:18:18 PM	18s	traffic	Succeeded	Original			3157f83c-75cc-4ece-ba1d-5ad5d4c1
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☐	Ingest_Traffic_API	11/23/2025, 10:18:00 PM	11/23/2025, 10:18:18 PM	19s	traffic	Succeeded	Original			8d0fea6f-caea-49f4-9b85-31b6d22ac
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☐	Ingest_Traffic_API	11/24/2025, 12:18:00 AM	11/24/2025, 12:18:20 AM	20s	traffic	Succeeded	Original			5eea0116-b455-4940-9e78-3672615l
☐	pipeline1	11/24/2025, 12:47:00 AM	11/24/2025, 12:49:48 AM	2m 49s	Daily_Files_Trigger	Failed	Original			67aaeea4-83da-4e62-96f9-f58a99eer
☐	Ingest_Weather_API	11/24/2025, 1:00:06 AM	11/24/2025, 1:00:28 AM	22s	Daily_Weather_Trigger	Succeeded	Original			ec648816-04b5-48da-a800-4e2c248z
☐	Ingest_Traffic_API	11/24/2025, 1:18:01 AM	11/24/2025, 1:18:17 AM	17s	traffic	Succeeded	Original			260e71e1-9c8f-47f1-965c-9d98c2e4!

Data Processing (Silver Layer)

Presenter: Ebrahim Ateff

Data Cleaning

Implemented PySpark logic to remove duplicates and handle missing values using mean imputation. This ensures high-quality input for the ML models.

Transformation

Weather: Exploded complex JSON arrays to align history.

Traffic: Flattened nested structures to calculate congestion indices.

Master Table: Joined all sources into silver.smartcity_master.

Data Processing (Silver Layer)

Presenter: Ebrahim Ateff

Data Cleaning Steps

1. **Mount & Load:** Read raw data from ADLS.
2. **Deduplication:** Used dropDuplicates().
3. **Validation:** Cast string timestamps to TimestampType.
4. **Null Handling:** Applied Mean Imputation for missing values.

Transformation

Flattening: Used explode() and arrays_zip() for nested JSON.

Feature Eng: Created peak_hour_flag (17:00-22:00) & weekend_flag.

Integration: Joined Energy + Weather + Traffic on timestamp.

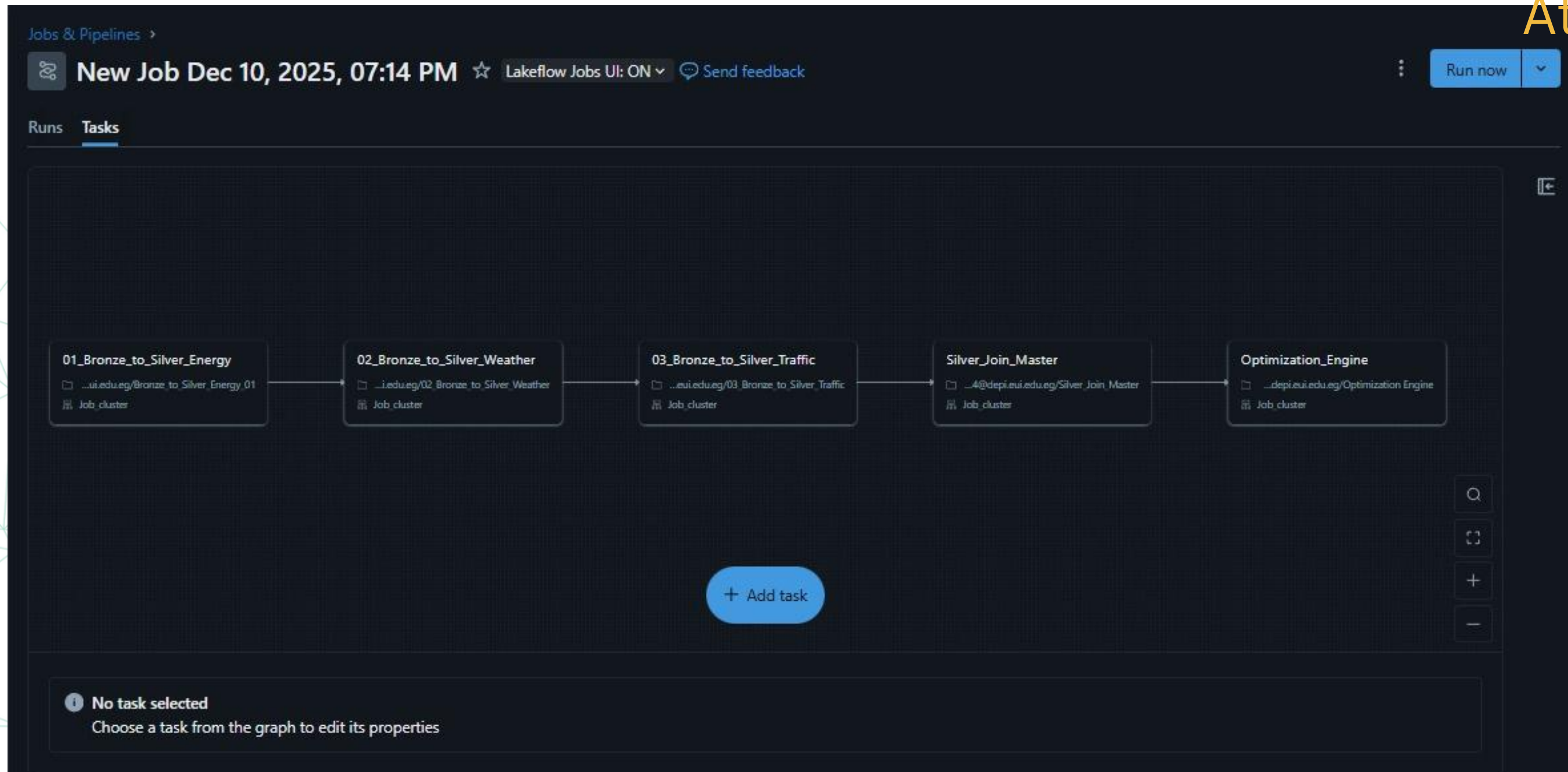
02_Bronze_to_Silver_Weather

03_Bronze_to_Silver_Traffic

Silver_Join_Master

Data Processing (Silver Layer)

Presenter: Ebrahim
Ateff



Forecasting & AI Models

Presenter: Abdullah Mohamed

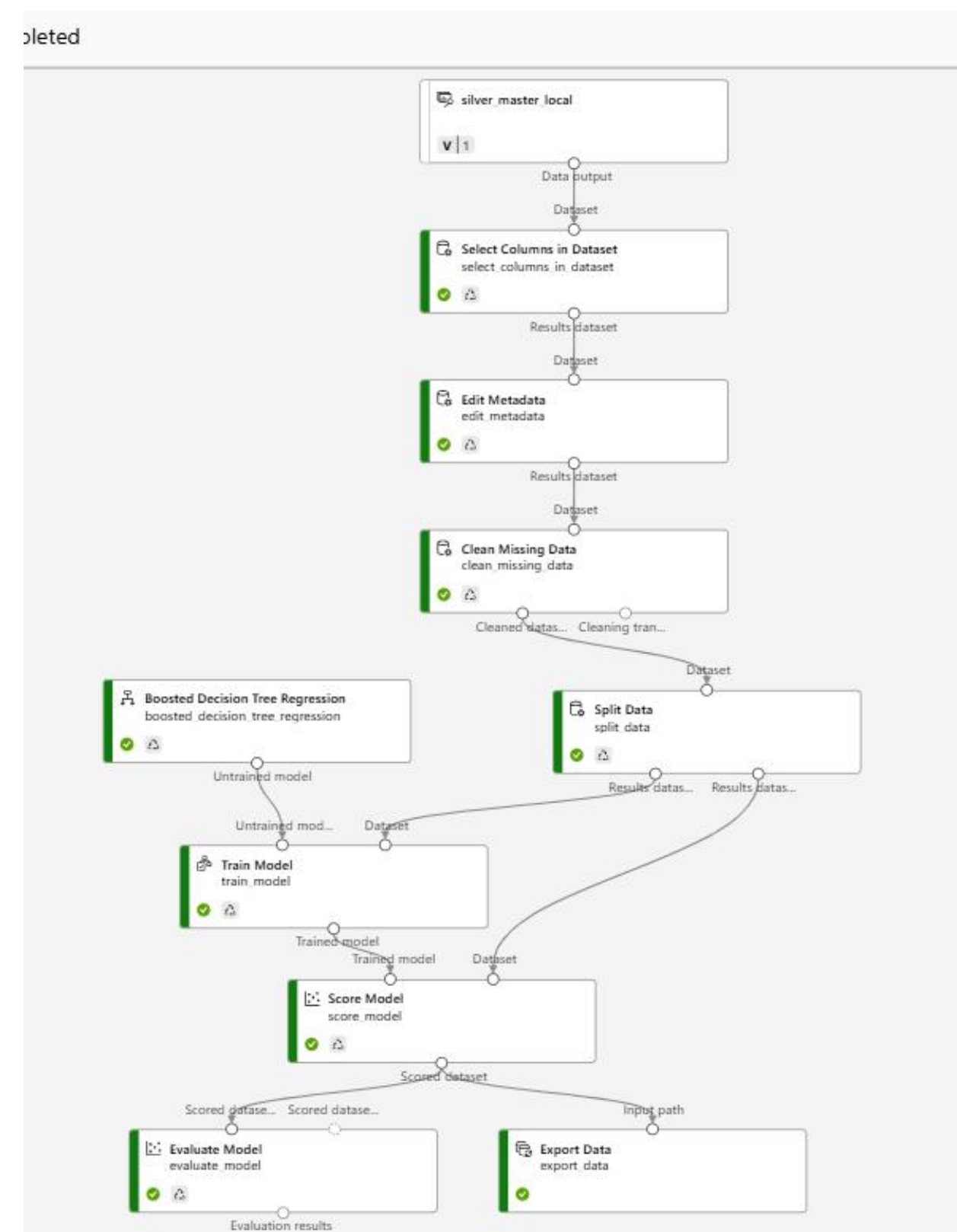
Microsoft Foundry | Azure Machine Learning

Scored_dataset

Rows 1,476 Columns 10

reading_time	region_id	energy_consumption	peak_hours_flag	month	weekday	feels_like_raw	humidity_raw	weather_index	Scored Labels
2025-12-06 12:00:00	r1	95.445655	0	12	7	18.108599	66	NaN	264.153552
2025-12-02 06:00:00	r4	305.054531	0	12	3	13.7	93	37.49	254.752747
2025-11-22 04:00:00	r2	318.715399	0	11	7	18.108599	66	NaN	245.430352
2025-12-23 07:00:00	r4	391.04826	0	12	3	18.108599	66	NaN	248.924946
2025-11-17 16:00:00	r3	87.713091	0	11	2	18.108599	66	NaN	250.108267
2025-11-24 11:00:00	r2	270.528355	0	11	2	18.108599	66	NaN	233.071701
2025-12-02 05:00:00	r4	153.416384	0	12	3	12.7	97	37.99	198.674467
2025-12-07 21:00:00	r2	150.013194	1	12	1	18.108599	66	NaN	259.55594
2025-01-07 10:00:00	r3	286.973644	0	1	3	18.108599	66	NaN	201.547484
2025-11-21 00:00:00	r1	82.030414	0	11	6	18.108599	66	NaN	248.107438
2025-12-12 22:00:00	r2	293.92459	1	12	6	18.108599	66	NaN	239.28352
2025-11-05 16:00:00	r1	196.426709	0	11	4	18.108599	66	NaN	226.313208
2025-01-02 18:00:00	r4	52.730539	1	1	5	18.108599	66	NaN	228.470875
2025-11-01 15:00:00	r3	170.319917	0	11	7	18.108599	66	NaN	234.612765
2025-01-01 03:00:00	r4	212.135493	0	1	4	18.108599	66	NaN	243.64687

deleted

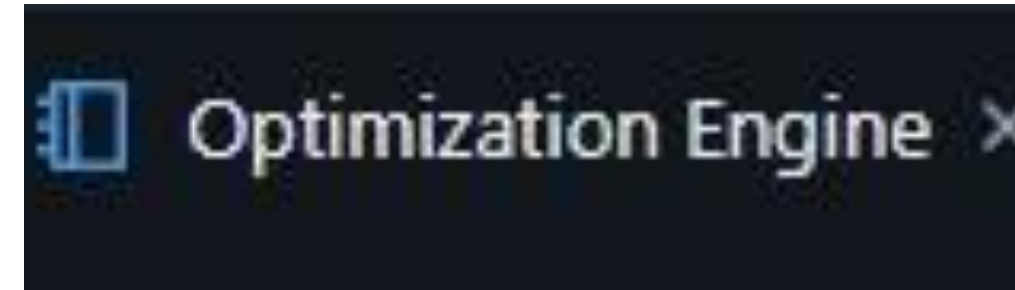


Forecasting & AI Models

Presenter: Abdullah Mohamed

Data Strategy: Split historical data into 80% Training and 20% Testing sets for validation.

 **Models:** Trained ARIMA and Prophet models on the Silver Master table to predict demand.



Gold Outputs:

- 🎯 - gold.energy_forecast: Hourly predictions.
- gold.risk_zones: Risk classification.
- gold.optimization: Load shifting logic.

Forecasting & AI Models

Presenter: Abdullah Mohamed

Creating the Serving Layer:

We utilized **Azure Synapse Serverless SQL Pools** to create a virtual layer over our data.

The script uses the **CREATE VIEW** command to define a schema for the ML output (energy_forecast).

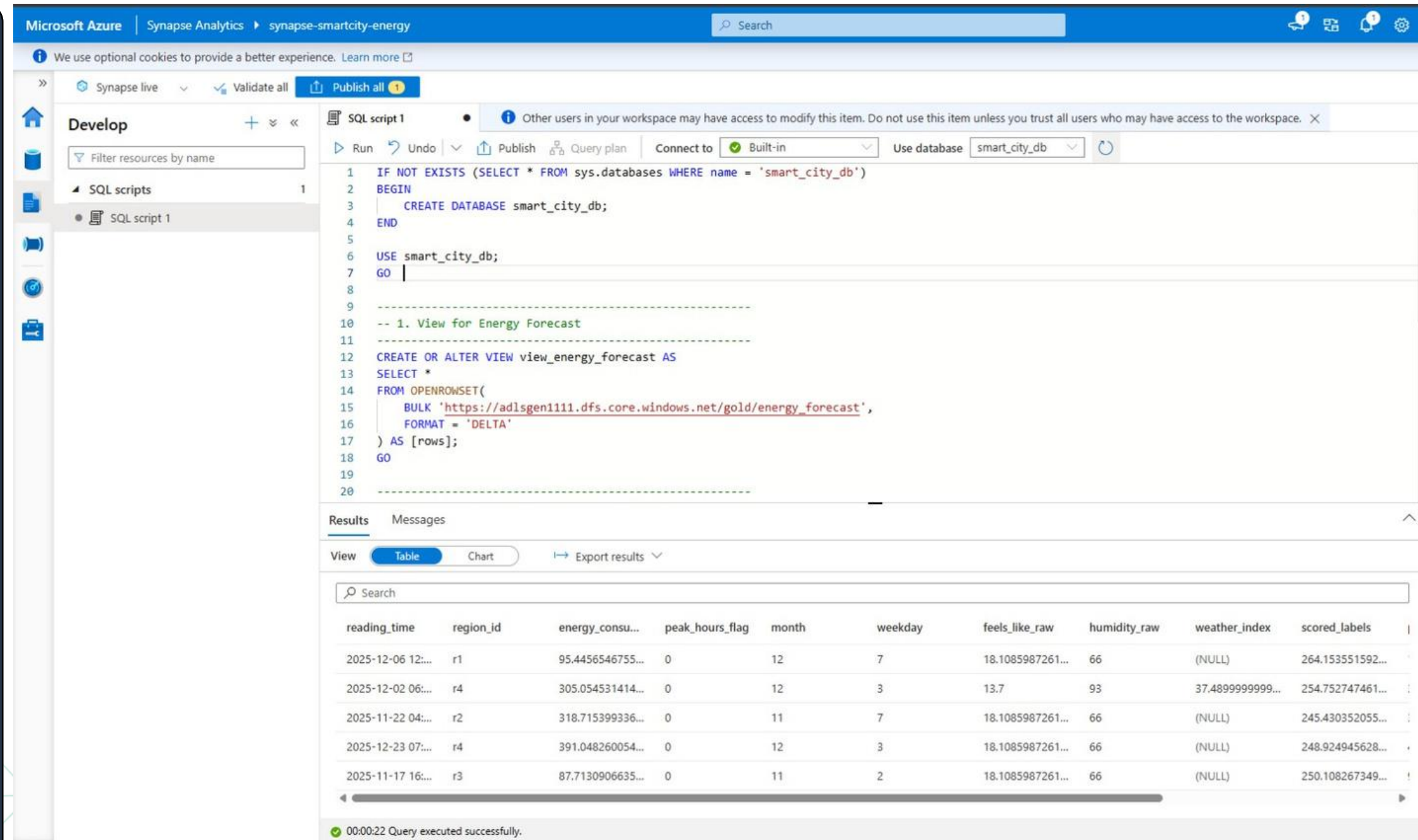
Data Virtualization (OPENROWSET):

We used the **OPENROWSET** function to query **Parquet/Delta files** directly from **ADLS Gen2** without moving the data.

This converts raw file paths into a structured SQL table format that Power BI can easily understand.

Seamless Integration:

This view acts as the connector. Power BI connects to this view via "DirectQuery" or "Import" mode to visualize the scored_labels (predictions) alongside actual_consumption.



The screenshot displays the Microsoft Azure Synapse Analytics interface. The top navigation bar shows "Microsoft Azure | Synapse Analytics | synapse-smartcity-energy". Below the navigation bar, there's a "Develop" section with a "Filter resources by name" input and a list of "SQL scripts" containing "SQL script 1". The main area shows the SQL script for "SQL script 1" with the following code:

```
1 IF NOT EXISTS (SELECT * FROM sys.databases WHERE name = 'smart_city_db')
2 BEGIN
3     CREATE DATABASE smart_city_db;
4 END
5
6 USE smart_city_db;
7 GO
8
9 -----
10 -- 1. View for Energy Forecast
11 -----
12 CREATE OR ALTER VIEW view_energy_forecast AS
13 SELECT *
14 FROM OPENROWSET(
15     BULK 'https://adlsgen1111.dfs.core.windows.net/gold/energy_forecast',
16     FORMAT = 'DELTA'
17 ) AS [rows];
18 GO
19
20 -----
```

Below the script, the "Results" section shows a table with the following columns: reading_time, region_id, energy_consu..., peak_hours_flag, month, weekday, feels_like_raw, humidity_raw, weather_index, and scored_labels. The table contains 6 rows of data, including timestamps, region IDs, energy consumption values, peak hours flags, months, weekdays, and various weather-related metrics.

At the bottom, a status bar indicates "00:00:22 Query executed successfully."

Power BI Integration

Presenter: Merna Sameh

 **Synapse Integration:** Leveraged Serverless SQL Views (view_energy_forecast) to expose Gold data securely to Power BI.

Key Visuals:

-  - Regional Energy Risk Analysis map.
-  - Forecast vs. Actual Consumption trends.

Impact: Provides decision-makers with a real-time view of grid health and optimization opportunities.



Live Application Status & Testing

✓ Production Ready

The system is fully deployed on Azure. Pipelines run daily, ingesting new data and updating the Power BI dashboard automatically.

Current State: **Beta / Live**
Data refreshes every 24 hours.

Testing Phases

- **Unit Testing:** Validated schema enforcement and data types within Databricks notebooks.
- **Integration Testing:** Successful end-to-end execution of ADF pipelines (Ingest -> Process -> Train -> Report).
- **Model Evaluation:** Forecasts evaluated using RMSE and MAPE metrics; successfully predicted historical peak events.

Project Deliverables



Source Code

Databricks Notebooks (Python), ADF Pipeline
JSONs, ML Training Scripts.



Documentation

Architecture Diagrams, Data Dictionary, API
Specs, Final Report.



Final Product

Interactive Power BI Dashboard connected to
live Synapse views.

Timeline: All milestones completed. Ready for final handover.

Project Team



Nagat Osama

Data Engineer
Ingestion Pipelines



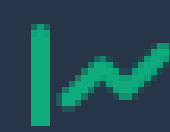
Ebrahim Ateff

Data Engineer
Processing &
Transformation



**Abdullah
Mohamed**

ML Engineer
AI Models &
Forecasting



Merna Sameh

Team Lead
BI & Visualization

Questions?

Thank you for your attention.

Team 4 | Smart City Energy Project

Thank You!

Questions?

GitHub: <https://github.com/nagatosamasaad>